Dataset Curation Flowchart

BV-BRC AMR Phenotype Database
Imipenem + Laboratory Method
n = 1,892
Resistant: 964 | Susceptible: 928

After mg/L units filter
(removed disk diffusion & missing values)
n = 930
Resistant: 363 | Susceptible: 567

Removed: 601 R, 361 S
(disk diffusion, missing MIC values)

After EUCAST 2023 breakpoints
(MIC ≥8 resistant, MIC ≤2 susceptible)
n = 876
Resistant: 362 | Susceptible: 514

Removed: 1 R, 53 S
(gray zone MIC=4, label mismatches)

After genome assembly download
(BV-BRC FASTA sequences)
n = 823
Resistant: 338 | Susceptible: 485

Removed: 24 R, 29 S
(no assembly available in BV-BRC)

After assembly quality control
(QUAST: N50 ≥10 kb, contigs ≤500,
length 3.5–4.5 Mb, GC 38–41%)
n = 805
Resistant: 330 | Susceptible: 475

Removed: 8 R, 10 S
(18 total failed QC thresholds)

After species verification
(removed non-A. baumannii)
n = 775
Resistant: 313 | Susceptible: 462

Removed: 17 R, 13 S
(30 Acinetobacter pittii, taxon 48296)

After phylogenetic outlier removal
(distance >3× IQR from ATCC 19606)
n = 772
Resistant: 313 | Susceptible: 459

Removed: 0 R, 3 S
(1 Acinetobacter sp., 2 extreme outliers)

Final Analysis Dataset
A. baumannii — Imipenem
n = 770
Resistant: 313 | Susceptible: 457

Retained
Final dataset
Excluded